

# Bt302 past papers solution or final term

## Define proto oncogene.

A proto-oncogene is a normal gene that can become an oncogene, either after mutation or increased expression.

Proto-oncogenes code for proteins that help to regulate cell growth and differentiation. Proto-oncogenes are often involved in signal transduction and execution of mitogenic signals, usually through their protein products. Upon activation, a proto-oncogene (or its product) becomes a tumor inducing agent, an oncogene.

## Define hypersensitivity.

Hypersensitivity (Immunological reaction)

refers to undesirable immune reactions produced by the normal immune system. Hypersensitivity reactions: When an immune response result in exaggerated OR in appropriate reactions harmful to the host, the term hypersensitivity OR allergy used. Hypersensitivity reactions: four types; based on the mechanisms involved and time taken for the reaction, a particular clinical condition (disease) may involve more than one type of reaction.

## Mechanism of hypersensitivity?

Mechanism of Type III Hypersensitivity: Antigens combines with antibody within circulation and form immune complex

- Wherever in the body they deposited
- They activate compliment system
- Polymorphonuclear cells are attracted to the site
- Result in inflammation and tissue injury

Of the mechanism of type III (immune-complex mediated hypersensitivity

## What you know about hypersensitivity type?

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- Wherever in the body they deposited
- They activate compliment system
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- Result in inflammation and tissue injury

## Innate and adaptive immunity.

**Inborn or innate immunity:** It is present at birth; This is our First Line Of Defense.

## Innate immunity

• If any invader penetrate the body's first line of defense mechanism is the innate immune system

- Components of innate immunity
- Epithelial barriers
- Cellular mechanisms
- Humoral mechanisms
- Role of innate immunity in stimulating adaptive immune response

## Adaptive Immune System (AIS)

AIS has two components.

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Humoral immunity is mediated by soluble antibody proteins produced by B lymphocytes that

neutralize and kill pathogens/toxins (20%-30%).

Cellular immunity is mediated by T lymphocytes which target cells infected by a pathogen

(60%-70%)

**Define antibody and five types of antibody.**

**Cytokine types .**

Cytokines:

- Cytokines are a diverse group of non-antibody proteins released by cells that act as intercellular mediators, especially in immune processes
- Low molecular weight soluble proteins (polypeptides) (Less than 30kD)
- produced in response to microbes and other antigens

## **Functional Categories of Cytokines:**

Cytokines classified according to their biologic actions into three groups:

### **1) Mediators and regulators of innate immunity**

- Produced by activated macrophages and NK cells in response to microbial infection
- they act mainly on endothelial cells and leukocytes to stimulate the early inflammatory response to microbes

### **2) Mediators and regulators of acquired immunity**

- Produced mainly by T lymphocytes in response to specific recognition of foreign antigens
- They include IL-2, IL-4, IL-5, IL-13, IFN, Transforming growth factor- $\beta$  (TGF- $\beta$ ) and lymphotoxin (TNF-  $\beta$ )

### **3) Stimulators of haematopoiesis**

- Produced by bone marrow, leukocytes
- Stimulate growth and differentiation of leukocytes
- Stem cell factors, IL-3, IL-7, GM-CSF

## **2. signal of cell mediated receptor?**

Cell signaling is part of any communication process that governs basic activities of **cells** and coordinates all **cell** actions. The ability of **cells** to perceive and correctly respond to their microenvironment is the basis of development, tissue repair, and immunity, as well as normal tissue homeostasis. Proteins responsible for detecting stimuli are generally termed receptors, although in some cases the term sensor is used.<sup>[1]</sup> The changes elicited by ligand binding (or signal sensing) in a receptor give rise to a **signaling cascade**, which is a chain of biochemical events along a signaling

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pathway. When signaling pathways interact with one another they form networks, which allow cellular responses to be coordinated, often by combinatorial signaling events.<sup>[2]</sup> At the molecular level, such responses include changes in the transcription or translation of genes, and post-translational and conformational changes in proteins, as well as changes in their location.

## 3. Three names of vaccines which are live attenuated?

Tuberculosis

Typhoid

Rotavirus

Varicella zoster (chickenpox)

Yellow fever

## 4. Explain the structure of B cell and T cell under microscope?

### B Cells

- BCRs bind intact antigens (like diphtheria toxoid, the protein introduced into your body in the DTP vaccine). These may be
    - soluble molecules present in the extracellular fluid;
    - intact molecules that the B cell plucks from the surface of antigen-presenting cells like macrophages and dendritic cells.
  - The bound antigen molecules are engulfed into the B cell by receptor-mediated endocytosis.
  - The antigen is digested into fragments
  - which are then displayed at the cell surface nestled inside a class II histocompatibility molecule.
  - Helper T cells specific for this structure (i.e., with complementary TCRs) bind the B cell and
  - secrete lymphokines that:
    - stimulate the B cell to enter the cell cycle and develop, by repeated mitosis, into a **clone** of cells with identical BCRs;
- [Link to discussion of clonal selection.](#)
- switch from synthesizing their BCRs as integral membrane proteins to a soluble version;
  - differentiate into **plasma cells** that secrete these soluble BCRs, which we now call **antibodies**.

### T Cells

The surface of each T cell also displays thousands of identical **T cell receptors (TCRs)**. There are two types of T cells that differ in their TCR:

- **alpha/beta** ( $\alpha\beta$ ) T cells. Their TCR is a heterodimer of an alpha chain with a beta chain. Each chain has a variable (V) region and a constant (C) region. The V

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regions each contain 3 **hypervariable regions** that make up the antigen-binding site. [\[Link\]](#)

- **gamma/delta** ( $\gamma\delta$ ) T cells. Their TCR is also a heterodimer of a gamma chain paired with a delta chain.

The discussion that follows now concerns alpha/beta T cells. Gamma/delta T cells, which are less well understood, are discussed at the end [\[Link\]](#).

The TCR (of alpha/beta T cells) binds a bimolecular complex displayed at the surface of **some other cell** called an [antigen-presenting cell](#) (**APC**). This complex consists of:

- a fragment of an antigen lying within the groove of a
- [histocompatibility molecule](#)

## 5. Mechanism of GVHR?

- ✓ Graft versus host reaction In some instance the graft tissue elicits an immune response against host antigen and that immune response is called graft versus host reaction
- ✓ Graft versus host reaction brings damage to host cells and host
- ✓ When grafted tissue has mature T cells, they will attack host tissue leading to GVHR.

Mechanism of GVHR:

- ✓ Graft lymphocytes aggregate in the host lymphoid organs
- ✓ Graft lymphocytes are stimulated by the host lymphocyte
- ✓ Stimulated lymphocytes of graft produce lymphokines
- ✓ Lymphokines activate host T- cell which produce polyclonal b-cell activation
- ✓ Activated b-cell react with the self antigens and cause damage to the host cell

## 6. Define antibodies ?and five antibody classis in mammals?

a blood protein produced in response to and counteracting a specific antigen.

Antibodies combine chemically with substances which the body recognizes as alien, such as bacteria, viruses, and foreign substances in the blood.

IgA (immunoglobulin A):

**Human immunoglobulin A (IgA)** is the second most common human immunoglobulin in serum. It is secreted in milk and is also the most prevalent Ig in secretions (e.g. tears, saliva and mucous).

IgA is resistant to digestion and can activate the complement pathway when aggregated. It should also be noted both subclasses of IgA (IgA1 and IgA2) bind fragment crystallization (Fc) receptor

IgD (immunoglobulin D):

Expressed on the surface of mature B cells, **human immunoglobulin D (IgD)** works with [IgM](#) in B cell development.

IgD is found in very low levels in serum and does not activate the complement pathway

IgE (immunoglobulin E):

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**Human immunoglobulin E (IgE)** is expressed on the surface of mature B cells. Human immunoglobulin IgE is the least abundant Ig in the serum and does not activate the complement pathway.

Fragment crystallization (Fc) receptors for IgE are found on eosinophils and IgE binds Fc receptors on mast cells and basophils even before interacting with antigen. As a result of its binding to basophils and mast cells, IgE is involved in allergic reactions. This happens when allergen is bound to IgE on cells and releases various pharmacological mediators which cause allergies.

IgG (immunoglobulin G):

**Human IgG (immunoglobulin G)** is expressed on the surface of mature B cells and is the most prevalent Ig in serum and the major Ig in extravascular spaces. **IgG1**, **IgG2** and **IgG3** are complement activators, with IgG3 being the strongest.

Human immunoglobulin subclasses IgG1 and IgG3 tend to strongly bind fragment crystallization (Fc) receptors, whereas subclasses IgG2 and IgG4 bind weakly. IgG is also the only human immunoglobulin to pass from mother to fetus to transfer immunity.

IgM (immunoglobulin M):

**Human IgM (immunoglobulin M)** is expressed on the surface of immature and mature B cells as monomers. IgM is the third most abundant human immunoglobulin.

IgM is also the first human immunoglobulin to be made by a fetus and virgin B cells which are challenged with antigen. IgM is a strong complement activator and agglutinator due to its pentameric structure and binds fragment crystallization (Fc) receptors.

## 7. Hypersensitivity?

## 8. Memory cell and Nk cell:

**Natural killer** cells (also known as **NK** cells, K cells, and **killer** cells) are a type of lymphocyte (a white blood cell) and a component of innate immune system. **NK** cells play a major role in the host-rejection of both tumours and virally infected cells.

### Memory cells:

a long-lived lymphocyte capable of responding to a particular antigen on its reintroduction, long after the exposure that prompted its production

### 1- What is antigen?

In immunology, an **antigen** is a molecule capable of inducing an immune response (to produce an antibody) in the host organism. Sometimes **antigens** are part of the host itself in an autoimmune disease. **Antigens** are "targeted" by antibodies.

### 2- Name two diseases that are caused by bacterial or viral infection used as vaccine?

Otitis media is the official name for an infection or inflammation of the middle ear. ... Respiratory tract infections include sore throat, bronchitis, sinusitis, and pneumonia. Bacteria or viruses may be responsible for respiratory tract infections. Tuberculosis is a

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type of bacterial lower respiratory tract infection.M

## 3- Name types of receptors of cytokines?

Type I receptor

Type II

Type III

Type IV

## 4- Write the function of spleen?

The **spleen** plays multiple supporting roles in the body. It acts as a filter for blood as part of the immune system. Old red blood cells are recycled in the **spleen**, and platelets and white blood cells are stored there. The **spleen** also helps fight certain kinds of bacteria that cause pneumonia and meningitis.

## 5- Clinical symptoms of graft versus host reaction?

Clinical manifestations of **acute** GVHD include a classic **maculopapular rash**; **persistent nausea** and/or emesis; abdominal cramps with **diarrhea**; and a rising serum bilirubin concentration

## 7- What is allergic contact dermatitis?

Allergic contact dermatitis:

- Cell-mediated immune response
- Results in an intensely irritating skin rash
- Triggered by chemically modified skin proteins that

the body regards as foreign

- Acellular, fluid-filled blisters develop in severe cases
- Can be treated with glucocorticoids

## 8- Write the function of lymph nodes?

**Lymph nodes** are major sites of B and T lymphocytes, and other white blood cells. **Lymph nodes** are important for the proper functioning of the immune system, acting as filters for foreign particles and cancer cells. **Lymph nodes** do not have a detoxification function, which is primarily dealt with by the liver and kidneys.

## 2. signal of cell mediated receptor?

Receptor-mediated endocytosis (RME), also called clathrin-mediated endocytosis, is a process by which cells absorb metabolites, hormones, other proteins – and in some cases viruses – by the inward budding of plasma membrane vesicles containing proteins with receptor sites specific to the molecules being absorbed (endocytosis).

## Cytokines and its types 5 mark:

- Cytokines are a diverse group of non-antibody proteins released by cells that act as intercellular mediators, especially in immune processes
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**types are in the functional types of cytokines.**

**Cytotoxic T cells 3 marks:**

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A **cytotoxic T cell** (also known as  $T_C$ , **cytotoxic** T lymphocyte, CTL, **T-killer cell**, **cytolytic T cell**, **CD8+T-cell** or **killer T cell**) is a **T lymphocyte** (a type of white blood **cell**) that kills cancer **cells**, **cells** that are infected (particularly with viruses), or **cells** that are damaged in other ways.

## cytokines function, mode of action and receptors types 10 mark

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**Three names of disease from attenuated vaccine 3 mark**

**Name of oxygen independent antibiotics 3**

**What are T cells and two classes of T cells 2 mark**

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A **T cell**, or **T lymphocyte**, is a type of lymphocyte (a subtype of white blood cell) that plays a central role in cell-mediated immunity. **T cells** can be distinguished from other lymphocytes, such as **B cells** and natural killer **cells**, by the presence of a **T-cell** receptor on the cell surface.

Types:

## Types

- Helper. T helper cells ( $T_H$  cells) assist other white blood cells in immunologic processes, including maturation of B cells into plasma cells and memory B cells, and activation of cytotoxic T cells and macrophages. ...
- Regulatory (suppressor) ...
- Natural killer T cell. ...
- Mucosal associated invariant. ...
- Gamma delta T cells.

Hypersensitive reaction type1 5 mark

- Commonly called allergy
- Mediated by IgE antibodies produced by plasma cells in response to stimulation of Th2 cells by an antigens.
- The antigens that stimulate it are called allergens (i.e. House dust, Pollens, Cosmetics, Insects, Clothing and Drug)
- Exposure may be ingested, inhalation, injection or direct contact.
- Type I hypersensitivity reactions can be systemic (e.g., systemic anaphylaxis) or localized to a specific target tissue or organ (e.g., allergic rhinitis, asthma).

## Bt302 What are interferons (2)

**Interferons** (IFNs) are a group of signaling proteins made and released by host cells in response to the presence of several pathogens, such as viruses, bacteria, parasites, and also tumor cells. ...**Interferons** are named for their ability to "interfere" with viral replication by protecting cells from virus infections.

## What are effector helper T cells (3)

**Effector cells** are the relatively short-lived activated **cells** that defend the body in an immune response. **Effector B cells** are called plasma **cells** and **secrete** antibodies, and activated **T cells** include cytotoxic **T cells** and helper **T cells**, which carry out cell-mediated responses. Antibodies have several modes of action: i) they can block ligand-receptor interactions; ii) cause cell lysis through activation of complement dependant cytotoxicity (CDC); iii) interact with Fc receptors on **effector** cells to engage antibody dependent cellular cytotoxicity; iv) signal for ingestion of a pathogen

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**Write names and functions of mhc proteins (3)**

**MHC:**

2 main structurally & functionally distinct classes of MHC proteins: class I MHC proteins, present foreign peptides to cytotoxic T cells, & class II present foreign peptides to helper T cells. The major histocompatibility complex (**MHC**) of genes consists of a linked set of genetic loci encoding many of the **proteins** involved in antigen presentation to T cells, most notably the **MHC** class I and class II glycoproteins (the **MHC** molecules) that present peptides to the T-cell receptor. The **importance of MHC proteins** is that they allow T cells to distinguish self from non-self. In every cell in your body, antigens are constantly broken up and presented to passing T cells. Without this presentation, other aspects of the immune response cannot occur

**write functions of spleen (3)**

**spleen:**

The **spleen** plays multiple supporting roles in the body. It acts as a filter for blood as part of the immune system. Old red blood cells are recycled in the **spleen**, and platelets and white blood cells are stored there. The **spleen** also helps fight certain kinds of bacteria that cause pneumonia and meningitis

**How b cells form igm antibody (5) Write the steps**

**Class Switching:**

B-Cell initially produces IgM (antibody's heavy chain is encoded by the first constant region gene, the  $\mu$  segment) molecules, which recognise specific antigenic determinant

**When B cell** becomes a plasma cell another deletion occurs in the cell's DNA, which position heavy-chain variable region gene next to a different constant region ( $\bar{U}$ ,  $\bar{I}$ , or  $\alpha$ ).

The First Step Interaction of Helper T Cells with Antigen on A Cells, and the Production of Lymphokines In this first step, growth and maturation factors

A clone of helper T cells can produce its own T

Lymphokines Produced by A Cells and the T-Cell Independent Activation of A Cells to a-factor Production The set of ly

**when mast cells are activated and leukocytes enter the site of injury (5)**

Mast cells :initiates inflammation by releasing biochemical mediators from preformed cytoplasmic granules and synthesizing other mediators in response to a stimulus

activation:

Physical injury, chemical agents, immunologic processes, and toll-like receptors

Chemical release in two ways: Degranulation and synthesis of lipid-derived chemical mediators

Histamine: released from mast cells

causes dilation of capillaries and retraction of endothelial cells lining the capillaries, which increases vascular permeability.

endothelial cells

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lines circulatory system

Release nitric oxide -> Vasodilation & Suppress inflammation

maintain normal blood flow by preventing spontaneous activation of platelets and members of the clotting system

expresses receptors that help leukocytes leave the vessel and retract to allow fluid to pass into the tissues.

platelets

interact with the coagulation cascade to stop bleeding

release a number of mediators that promote and control inflammation

neutrophil & phagocyte

early inflammatory response, they move to the inflammatory site by chemotaxis

exits the circulation by diapedesis through the retracted endothelial cell junctions and moves to the inflammatory site by chemotaxis

eosinophils

release products that control the inflammatory response and are the principal cell that kills parasitic organisms

macrophage

predominant phagocytic cell in the late inflammatory response

responsive to cytokines

promotes wound healing

Dendritic cells

connect the innate and acquired immune systems by collecting antigens at the site of inflammation and transporting them to sites

ie lymph nodes, where B and T cells live

Phagocytosis

process for the elimination of pathogens and foreign debris

Steps:

recognition and attachment,

engulfment,

formation of a phagosome and phagolysosome,

destruction of pathogens or foreign debris

engulf microorganisms and enclose them in phagocytic vacuoles

toxic products and degradative lysosomal enzymes kill and digest the microorganisms.

Opsonins

antibody and complement component C3b

coat microorganisms and make them more susceptible to phagocytosis by binding them more tightly to the phagocyte

Local manifestations of inflammation

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result of the vascular changes associated with the inflammatory process, including vasodilation and increased capillary permeability

symptoms: redness, heat, swelling, and pain.

Systemic Manifestations of Acute Inflammation

fever and increases in levels of circulating leukocytes and plasma proteins.

Systemic Manifestations of Inflammation

Fever

Caused by exogenous and endogenous pyrogens

Act directly on the hypothalamus

Leukocytosis

Increased numbers of circulating leukocytes

Shift to the left

Increased plasma protein synthesis

Acute-phase reactants

C-reactive protein (CRP), erythrocyte sedimentation rate (ESR)

chronic inflammation

lasts 2 weeks or longer

characterized by a dense infiltration of lymphocytes and macrophages

formation of a granuloma: isolate the infection to protect against tissue damage

Causes: High lipid and wax content of a microorganism (Mycobacteria TB)

Ability to survive inside the macrophage (Leprosy, syphilis)

Toxins (Chemicals, particulate matter, or physical irritants)

Resolution or regeneration return of tissue to nearly normal structure and function.

Repair healing by scar tissue formation Scar tissue Composed primarily of collagen to restore the tensile strength of the tissue

cannot carry out the physiologic functions of destroyed tissue

Debridement Cleaning up the dissolved clots, microorganisms, erythrocytes, and dead tissue cells. Healing Filling in the wound Sealing the wound (epithelialization) Shrinking the wound (contraction) healing by primary intention. Damaged tissue proceeds to resolution if little tissue has been lost or injured tissue is capable of regeneration healing by secondary intention.

Tissues that sustained extensive damage or those incapable of regeneration heal by the process of repair resulting in the formation of a scar resolution & repair: reconstructive phase wound begins to heal resolution & repair: maturation phase healed wound is remodeled. Dysfunctional wound healing can occur as a result of abnormalities in either the inflammatory response or the reconstructive phase of resolution and repair.

Dysfunction during inflammatory response:

Wound sepsis

Hypovolemia

Hypoproteinemia

Anti-inflammatory steroids

Dysfunction during reconstructive phase:

Impaired collagen matrix assembly = Keloid scar, Hypertrophic scar

Impaired epithelialization = Anti-inflammatory steroids, hypoxemia, and nutritional

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deficiencies Impaired contraction = Contracture three layers of human defense barriers, innate immunity (inflammatory response), and adaptive immunity. Physical and mechanical barriers first barriers of defense prevent damage to the individual and prevent invasion by pathogens skin & mucous membranes biochemical barriers Antibacterial peptides in mucous secretions, perspiration, saliva, tears, and other secretions. The Inflammatory Process: Inflammation is a rapid and nonspecific protective response to cellular injury from any cause. occurs only in vascular tissue macroscopic hallmarks of inflammation redness, swelling, heat, pain, and loss of function of the inflamed tissues microscopic hallmarks of inflammation accumulation of fluid and cells at the inflammatory site.

## What are antigens?

While **antigens** are the "target" of antibodies, immunoglobulin-binding proteins "attack" antibodies. . . . T-independent **antigen** – Polysaccharides (usually) that stimulate B cells directly. Immunodominant **antigens** – **Antigens** that dominate (over all others from a pathogen) in their ability to produce an immune response

## Define humoral and cellular immunity?

**Humoral immunity** or humoral **immunity** is the aspect of **immunity** that is mediated by macromolecules found in extracellular fluids such as secreted antibodies, complement proteins, and certain antimicrobial peptides. ... Its aspects involving antibodies are often called antibody-mediated **immunity**.

**Cell-mediated immunity** is an **immune** response that does not involve antibodies, but rather involves the activation of phagocytes, antigen-specific cytotoxic T-lymphocytes, and the release of various cytokines in response to an antigen

## Function of effector b cells?

## Function of spellen?

## What are the activation of cytotoxic T cells?

Helper **T cells** are arguably the most important **cells** in adaptive immunity, as they are required for almost all adaptive immune responses. They not only help **activate B cells** to secrete antibodies and macrophages to destroy ingested microbes, but they also help **activate cytotoxic T cells** to kill infected target **cells**

## B cell and t cell structure:

Lymphocytes are one of the five kinds of **white blood cells** or **leukocytes**), circulating in the blood. [\[More\]](#)

Although mature lymphocytes all look pretty much alike, they are extraordinarily diverse in their functions. The most abundant lymphocytes are:

- B lymphocytes (often simply called **B cells**) and
- T lymphocytes (likewise called **T cells**).

B cells are produced in the **bone marrow**.

The precursors of T cells are also produced in the bone marrow but leave the bone marrow and mature in the **thymus**

Each B cell and T cell is **specific** for a particular **antigen**. What this means is that each is able to **bind to** a particular molecular structure.

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The specificity of binding resides in a **receptor** for antigen:

- the B cell receptor (**BCR**) for antigen and
- the T cell receptor (**TCR**) respectively.
- They have a unique binding site.
- This site binds to a portion of the antigen called an [antigenic determinant](#) or **epitope**.
- The binding, like that between an [enzyme](#) and its [substrate](#) depends on complementarity of the surface of the receptor and the surface of the epitope.
- The binding occurs by [non-covalent forces](#) (again, like an enzyme binding to its substrate).
- Successful binding of the antigen receptor to the epitope, **if accompanied by [additional signals](#)**, results in:
  - stimulation of the cell to leave  $G_0$  and enter the [cell cycle](#).
  - Repeated [mitosis](#) leads to the development of a [clone](#) of cells bearing the same antigen receptor;

## AIS activation?

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AIS has two components.

Humoral immunity is mediated by soluble antibody proteins produced by B lymphocytes that

neutralize and kill pathogens/toxins (20%-30%).

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### IS Activation

IIS Phagocytic cells have receptors that have affinity for certain pathogen-associated molecules

IIS programmed to be activated by specific pathogen molecules (Toll Receptors)

IIS activates AIS

Adaptive Immune Response AIR is highly specific to the pathogen that induced it.

There are  $2 \times 10^{12}$  lymphocytes in the human body, making the immune system comparable in cell mass to the liver or brain.

### Cells of immune system:

- B Cells.
- Dendritic Cells.
- Granulocytes.
- Innate Lymphoid Cells (ILCs)
- Megakaryocytes.
- Monocytes/Macrophages.
- Myeloid-derived Suppressor Cells (MDSC)

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- Natural Killer (NK) Cells.

**Cytotoxic T Cell ?**

**Lymph node function ?**

**Lymph nodes** are major sites of B and T lymphocytes, and other white blood cells. **Lymph nodes** are important for the proper functioning of the immune system, acting as filters for foreign particles and cancer.

**Difference between B Cells and at Cells under electron Microscope. 2**

**What is the function of Fc antibody in IgM 2**

placenta cells that are in contact with mothers blood have Fc receptors that are use to grab IgG and pass it to the fetus.

**MHC Genes are represented for? Which cells are included in it and which protiens encode the**

**MHC genes** are located within the human major histocompatibility complex (**MHC**) and code for different members of the class I and the class II human leukocyte antigen (HLA) families. ... The open and shaded boxes **represent** pseudogenes and coding **genes** respectively. MHC proteins.

**MHC. 3**

**Functions of Thymus 3**

The **thymus** gland, despite containing glandular tissue and producing several hormones, is much more closely associated with the immune system than with the endocrine system. The **thymus** serves a vital **role** in the training and development of T-lymphocytes or T cells, an extremely important type of white blood cell. The **thymus gland** is located in the neck above the heart. The **function** of the **thymus gland** is to generate mature T lymphocytes (white blood cells that help the immune system fight off illness).

